Financial Modeling

Class 4 — Risk Management and Backtesting

In-class session

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Where we are in the course

- The construction block is complete. Before the strategies block, this class covers two tools that every strategy requires: **risk management** and **backtesting**
- Both apply directly to this week's work: the round-1 report must contain a backtest of the strategy
- AMC round 1 begins next week with the presentations

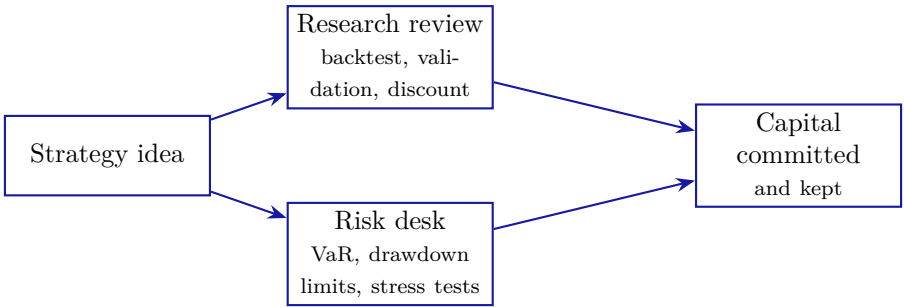
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Today: risk management and the reliability of backtests

- Two practices distinguish professional investment management: the control of losses before they accumulate, and a critical attitude towards one's own simulations
- Risk management addresses what volatility does not measure: the left tail of the return distribution, where the largest losses occur
- Backtesting is a separate question: a simulation of the past overstates performance. The class discusses by how much, and which procedures correct for it
- Both questions apply to the backtest required in the round-1 report

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The two control functions of a fund



A strategy is not allocated capital before it has been reviewed by both functions. In the AMC, each group performs both roles: the round-1 report contains the backtest and its risk measures.

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Exercise: drawdown control, a reactive rule

Sheet *Drawdown control*: S&P 500 and risk-free monthly returns, 2000–2009.

1. **The index.** Each month: value of 100 invested, highest value so far, drawdown (value divided by the highest value so far, minus one). Note the largest drawdown.
2. **The rule.** Each month set the exposure to the index $w_t = \min(1, \max(0, (15\% - DD_{t-1})/7.1\%))$, where DD_{t-1} is the drawdown of *your portfolio* at the end of the previous month, 15% the maximum drawdown the clients accept, 7.1% the monthly VaR of the index. The rest is in cash: portfolio return $= w_t \times$ index return $+(1 - w_t) \times$ risk-free rate.
3. **The portfolio.** As in step 1: value, highest so far, drawdown. Note the largest drawdown.

Report the portfolio's largest drawdown, in percent. 12 minutes, in pairs.

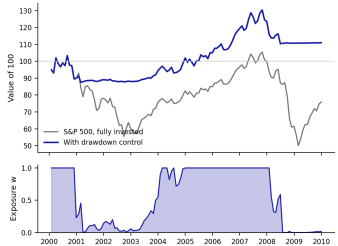
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The solution: the loss stays at the budget

2000–2009	Index	Rule
Maximum drawdown	52.6%	**15.3%**
Value of 100, end of 2009	75.9	111.0
Volatility (annual)	16.1%	7.9%
Months with exposure < 1	0 of 120	67 of 120
Average exposure	1.00	0.55



- The rule cuts the exposure *before* the budget is reached and adds it back as the drawdown recedes: a loss of 15% against 53%, a mechanical defense against the disposition effect
- The cost: once the budget is spent, the exposure stays near zero until the portfolio is back at its previous peak (2001–2003, after 2008). A variant restarts the budget every January: at most 17.6% within a year, but the yearly losses add up to 36.7% over the decade

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Vote: which protection works in which conditions?

Consider a slow, protracted bear market, such as 2000–2002.

Which strategy protects the portfolio best?

out-of-the-money puts trend following holding gold

Trend following: a rule applied to each asset (equity, bond, currency and commodity futures) that goes long when the asset's own return over the past 1 to 12 months is positive and short when it is negative; positions are revised monthly.

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Risk-mitigating strategies in drawdowns

Peak	Trough	Months	Fall, %	Index put	Trend	10-y Treasury	Gold	QMJ
Mar 1920	Aug 1921	17	−26		10	−4		
Aug 1929	Jun 1932	34	−84		**54**	−1		
Jan 1937	Mar 1938	13	−50		−8	6		
May 1946	Nov 1946	6	−22		4	−1		
Jul 1956	Dec 1957	17	−18		−1	−1		1
Nov 1961	Jun 1962	7	−23		0	0		0
Jan 1966	Sep 1966	8	−18		6	−2		2
Nov 1968	Jun 1970	19	−37		37	−13		34
Dec 1972	Sep 1974	21	−50		**123**	−11	129	5
Nov 1980	Jul 1982	20	−35		18	−6	−59	13
Jun 1983	Jul 1984	12	−15		2	−6	−28	17
Aug 1987	Nov 1987	3	−31	**23**	−6	0	7	4
Aug 1989	Oct 1990	14	−18	−6	11	−1	−4	29
Apr 1998	Aug 1998	4	−15	11	13	4	−12	12
Mar 2000	Sep 2002	30	−49	**11**	**52**	23	5	96
Oct 2007	Feb 2009	16	−52	**31**	15	14	14	53
Apr 2011	Sep 2011	5	−16	9	−8	10	4	20
Dec 2019	Mar 2020	3	−20	**27**	−2	10	4	4

Ilmanen (2022), Table 13.1, data from AQR. Returns over each episode, in percent; the S&P 500 in excess of cash.

- **Red:** monthly puts 5% out of the money pay when the fall is concentrated in a few months: 1987, 2008, 2020. In the slow decline of 2000–02 (**orange**) the monthly moves rarely reached the strike: 11% for a 49% fall
- **Blue:** trend following has time to turn short in protracted bear markets: 1929–32, 1973–74, 2000–02. Puts cost a premium every month, and the payouts are rare

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Recap 1 of 2: the toolbox of risk management

- **Forward-looking**: diversification (removes idiosyncratic risk), hedging (offsetting positions, no risk premium is earned), insurance (premium paid for downside protection)
- **Reactive**: drawdown control, the rule applied in the exercise
- **Measurement**: volatility as the starting point; **VaR** and **Expected Shortfall** for the left tail; **stress tests** for events too rare to estimate (September 11, Lehman, COVID)
- The AMC reports must include these measures for the strategy: volatility, maximum drawdown, and VaR

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Vote: the arithmetic of data mining

A researcher tests 20 variations of a strategy. Suppose all true alphas are zero.

At the 5% significance level, how many variations appear significantly profitable?

none about 1 about 4

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Exercise: what do you report?

A group's backtest shows a Sharpe Ratio of **1.4**, obtained after about **25 signal variations**, with all parameters tuned on the full sample.

Which number should the group report?

(a) 1.4, the value produced by the backtest (b) about 0.7, applying the discount for data mining (c) none, until the strategy is validated on data that was not used for tuning

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The answer, and the rules for the AMC backtests

- Both (b) and (c) are defensible; (a) is not. The rule of thumb in the notes: **divide the backtested Sharpe Ratio by 2**, with a larger discount when the strategy has more tunable parameters
- **Avoidable biases**, to be eliminated: a biased universe (historical constituents, not current survivors), look-ahead bias in the signal (no trading on information before it was available), in-sample-only fitting
- **Out-of-sample validation**: tune the parameters on one period, validate on another; repeated alternation between the two contaminates the validation data
- Quantitative funds monitor the number of simulations their analysts run. Each group should count its trials and report the count

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Exercise: a look-ahead error in a backtest

Sheet *Look-ahead*: daily US market returns, 1926–2026. The rule (Lo, 2015): each day, exposure $\lambda_t = \min(1.3,\ 17\%/\hat{\sigma}_{t-1})$, with $\hat{\sigma}_{t-1} = \sqrt{252}\times$ the standard deviation of the previous 21 daily returns. A fraction λ_t of the portfolio is in the index, the rest $1-\lambda_t$ in cash at the risk-free rate rf_t; when $\lambda_t > 1$ the extra exposure is borrowed at rf_t (negative cash). Strategy return $\lambda_t\, r_t + (1-\lambda_t)\, rf_t$. Row 23 is an example.

1. **The rule as it can be traded:** fill columns D to G down to the third-to-last row (volatility, exposure, strategy return, growth of \$1)
2. **The same rule with a three-day error:** columns H to K, the window shifted to $r_{t-18}, \ldots, r_{t+2}$, so that the estimate includes today's return and the next two
3. **Column L**, the growth of \$1 in the index; then the summary block: the growth of \$1 at the end for the three versions

Report the ratio shifted over correct. 10 minutes, in pairs.

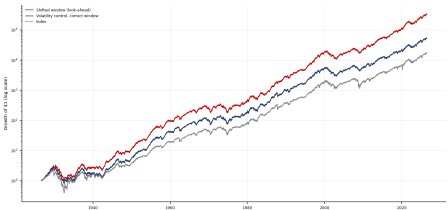
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The results: one index in a formula

	Index	Lagged	Look-ahead
Mean (ann.)	10.8%	11.6%	13.2%
Volatility	17.1%	15.2%	14.3%
Sharpe Ratio	0.46	0.56	0.71
Growth of $1	$17,075	$52,007	$322,926



- The shifted window multiplies the terminal wealth by **6**, yet the rule cannot be traded: on a day the market falls, the estimate already contains the loss and the exposure is cut *before* it. In real time, the loss comes first
- The error is one index in a formula. The manager can avoid it; the investor who reads the backtest cannot detect it

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260,400 rules, and three remedies

Even without look-ahead bias, the rule has four parameters. Over a grid of plausible values (window 2 to 125 days; cap 1.00 to 2.00 in steps of 0.05; 1 to 10 lags in the volatility estimate; transaction cost 1 to 10 bp), $124 \times 21 \times 10 \times 10 = 260{,}400$ versions of the same rule: the best one looks good even if none has value.

Lo's three remedies

1. **Skepticism.** Every record, simulated or live, contains luck; how much of the estimated alpha is noise?
2. **Additional information.** Look beyond the return series: a stock picker who does well only in rising markets delivers market exposure, not skill; an allocator should show a performance attribution
3. **Live out-of-sample evaluation.** Follow the strategy for a year or two on new data, after the backtest

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Exercise: momentum and volatility timing

Sheet *Momentum*: French's momentum factor (winners minus losers), the market excess return, the T-bill rate, monthly since 1927, and the realised volatility of momentum over the previous six months. Momentum is a long-short portfolio: do not subtract the risk-free rate.

1. **Raw momentum:** annualised mean, volatility, Sharpe Ratio, t-statistic of the mean
2. **Its alpha:** regress momentum on the market excess return (the Regression tool, or `INTERCEPT`, `SLOPE`, `STEYX`): alpha, beta, residual volatility; information ratio = alpha / residual volatility
3. **Volatility timing** (Barroso and Santa-Clara, 2015): scale the position by $w_t = 12\%/\hat{\sigma}_{t-1}$; the same statistics for $w_t \times$ momentum
4. If time allows: value of 1 invested and maximum drawdown, raw and timed

Report the Sharpe Ratio of the volatility-timed momentum. 12 minutes, in pairs.

Report your number on Wooclap, before the solution is shown:

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Solution: momentum and volatility timing

Momentum factor, 1927–2026 (1,192 months)	Raw	Volatility-timed
Mean return, annualised	7.3%	15.4%
Volatility, annualised	16.3%	17.8%
Sharpe Ratio	0.45	**0.87**
Alpha on the market, annualised (t-statistic)	9.8% (6.3)	16.6% (9.3)
Beta on the market	−0.30	−0.14
Information ratio	0.64	**0.94**
Maximum drawdown	−78%	−43%
2009, the momentum crash	−53%	−17%

- Over a century the alpha of momentum is beyond doubt ($t = 6.3$); the risk is in the crashes (1932, 2009), when the losers rebound after months of high volatility
- Scaling by the inverse of recent volatility takes less risk exactly then: the Sharpe Ratio nearly doubles, the worst drawdown halves. Ignored by the backtest: leverage (w_t above 1 in most months) and trading costs

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Implications for the round-1 report

- A backtested average return is an estimate with a large standard error; report it together with its **t-statistic**
- Report the **risk** of the backtested strategy, not only its return: volatility, maximum drawdown, and the 5% VaR
- State **how the backtest was run**: universe, signal, rebalancing frequency, sample period, and which part of the data was held out for validation
- Apply and state the **discount for data mining**

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Assignment: the backtest protocol (round-1 report)

Include a **backtest protocol** section in the round-1 report:

1. **Design**: universe, the rule that determines the weights, rebalancing frequency, and sample period of the backtest
2. **Validation**: which part of the data was used for tuning and which was held out; the number of variations the group tried
3. **Risk**: the volatility, maximum drawdown, and 5% VaR of the backtested strategy
4. **The discount**: the backtested Sharpe Ratio, the discounted value the group considers realistic, and the justification for the discount

Graded as complete or incomplete with the round-1 report. The guidelines of the round require a backtest with monthly rebalancing; this section documents how it was carried out.

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Next class

- **Discretionary equity investing**: value, quality, catalysts, activism, and equity valuation, which produces the alphas required by the Treynor–Black model
- The reading and the exercise workbook are on iCorsi
- AMC round 1 begins before the next class. The investment policy page, the comparison with the robust approaches, and today's backtest protocol are all part of the round-1 report. The deadline is in the guidelines of the round

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